

RGIPT

Subject: Ai

Class: 12

Time Allowed: 45 minutes

Maximum Marks: 10

All questions are compulsory unless stated otherwise.

Name: _____

Roll Number: _____

Class: 12 Section: _____

Section A: Multiple Choice Questions

Choose the correct option. Each question carries 1 mark.

1. [Easy] What does Machine Learning primarily involve? A) Writing explicit rules B) Showing many examples to a machine C) Hard-coding intelligence D) Using only explicit knowledge [1 Marks]
2. [Easy] Which combination of factors led to the AI explosion around 2012? A) Larger datasets only B) GPUs only C) Deep Learning algorithms only D) Massive data, powerful GPUs, and deep learning algorithms [1 Marks]
3. [Moderate] In word-embedding arithmetic, "King - Man + Woman" equals: [1 Marks]
4. [Moderate] Which architecture processes all words in a sentence simultaneously using attention? A) RNN B) CNN C) Transformer D) Decision Tree [1 Marks]

Section B: Short Answer Questions

Answer briefly. Each question carries 2 marks.

1. [Moderate] Explain why rule-based AI struggles with tasks like recognizing sarcasm or idioms. [2 Marks]
2. [Challenging] Describe the role of the attention mechanism in a Transformer model. [2 Marks]
3. [Easy] What are the three ingredients that converged around 2012 to enable large-scale language models? [2 Marks]

End of Question Paper

Answer Key:

1. B

2. D

3. Queen

4. C

5. Rule-based AI relies on explicit if-then rules, which cannot capture the nuanced, context-dependent meanings of sarcasm or idioms; the required rules would be countless and still miss subtle variations.
6. Attention lets every word in a sentence weigh and incorporate information from all other words simultaneously, enabling the model to focus on the most relevant context regardless of distance, eliminating forgetting and improving parallel processing.
7. Massive internet data, powerful GPUs for compute, and deep-learning algorithms (especially Transformers).